



PRAYCG Master Comprehensive Analysis Module Guide

Current integrated explanation of the acquisition, quality-control, primary, secondary, exploratory, confound-defense, and review modules

Central boundary

This guide explains operational indices and software gates. It does not establish consciousness, narrative reception, memory formation, clinical state, cellular mechanisms, thermodynamic energy transfer, or metaphysical claims.

Layer	Current role
Infrastructure/QC	Establish identity, timebase, coverage, artifacts, and applicability before interpretation.
Primary candidate path	Timing/QC + stimulus/confound gates -> A-MRED or MRED-Peak/Resolution at locked anchors.
Secondary summaries	Describe immersion, task tradeoffs, decoder availability, autonomic context, and baseline change.
Exploratory candidates	Generate falsifiable development quantities without altering primary eligibility.
Confound defenses	Challenge sensory, structural, gaze, order, afterglow, respiration, and task explanations.

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PART I

Data identity, synchronization, and core quality

These modules determine whether the data represent the intended run, conditions, channels, and timebase before scientific indices are calculated.

1. Run metadata and media identity

REQUIRED PROVENANCE

Tier: Primary QC Verifies participant/session labels, protocol version, branch videos, cue schedule, anchor file, hash identity, and configured durations.

What the module is trying to prevent

This module exists primarily to prevent a false-positive interpretation. It asks whether the run, timing, stimulus, or signal quality is strong enough to allow any downstream claim. If this layer fails, the correct output is not a weaker positive result; it is a downgraded or non-interpretable run.

Inputs and assumptions

Runner event log, run_config_media_selection.json, media hashes, cue schedule summary.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Parse setup markers and sidecar JSON. Confirm Target/Override hash match. Record cue count, expected sum, baseline/washout durations, anchor-file status, channel-map status, and PRAYCG version.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

run_identity_summary.csv, module registry rows, interpretation JSON fields.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

A valid run should show the selected media and cue schedule unambiguously. Target and Override should be hash-identical when the task differs only by instruction.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Missing config, Target/Override mismatch, unrecorded anchor file, absent cue schedule, or unknown protocol version downgrades claims.

Safe public wording

Recommended wording

This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

2. Stream inventory and timebase reconstruction

REQUIRED SYNCHRONIZATION

Tier: Primary QC Determines which LSL streams are present and whether physiology can be aligned to event markers.

What the module is trying to prevent

This module exists primarily to prevent a false-positive interpretation. It asks whether the run, timing, stimulus, or signal quality is strong enough to allow any downstream claim. If this layer fails, the correct output is not a weaker positive result; it is a downgraded or non-interpretable run.

Inputs and assumptions

XDF streams, StasisMarkers, OpenBCIStatusMarkers, sample timestamps.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Inventory stream names, sample rates, channel counts, durations, time gaps, and marker intervals. When raw timestamps are unreliable, reconstruct the OpenBCI timebase using sample-index and status markers.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

stream_inventory_corrected.csv, phase_coverage.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Successful reconstruction supports phase-sliced features even when raw XDF timestamps are imperfect.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Missing EEG stream, major dropout, unsolved timebase inconsistency, or poor phase coverage prevents strong interpretation.

Safe public wording

Recommended wording

This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

3. ALS/PT19 display-timing QC

TIMING EVIDENCE

Tier: Primary QC Checks whether physical screen transitions were detected at Control, Target, and Override starts.

What the module is trying to prevent

This module exists primarily to prevent a false-positive interpretation. It asks whether the run, timing, stimulus, or signal quality is strong enough to allow any downstream claim. If this layer fails, the correct output is not a weaker positive result; it is a downgraded or non-interpretable run.

Inputs and assumptions

ALS_PT19_Timing stream or OpenBCI analog AUX, branch start markers, cue/media timing manifest.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Detect black-white-black or timing-square pulses around branch-start markers. Estimate onset offset, duration, clipping, and pulse presence.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

als_start_pulse_qc.csv, ALS timing summary.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

A clean pass strengthens display-onset alignment. ALS belongs to external input $u(t)$, not biology.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Missed or ambiguous pulses downgrade anchor timing and endpoint claim grade.

Safe public wording

Recommended wording

This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

4. Channel map and ROI audit

REQUIRED MAPPING AUDIT

Tier: Primary QC Checks whether channel-to-electrode mapping is confirmed before ROI-based interpretation.

What the module is trying to prevent

This module exists primarily to prevent a false-positive interpretation. It asks whether the run, timing, stimulus, or signal quality is strong enough to allow any downstream claim. If this layer fails, the correct output is not a weaker positive result; it is a downgraded or non-interpretable run.

Inputs and assumptions

channel_map.csv, event markers, run configuration.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Parse PRAYCG16 channel map markers and confidence flags. Mark ROI use as allowed, cautioned, or blocked.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

channel_map_summary.csv or module registry field.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

LOCKED channel map allows frontal/parietal/occipital/temporal ROI labels to be used cautiously.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Unconfirmed channel map blocks ROI-specific claims even if global features can still be analyzed.

Safe public wording

Recommended wording

This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

5. Feature table generation

SHARED FEATURE LAYER

Tier: Primary engine Creates the time-resolved table used by all downstream modules.

What the module is trying to prevent

This module supports the core analysis chain. It should be interpreted only in relation to timing, artifact, Control, Override, and stimulus-regressor checks.

Inputs and assumptions

Cleaned/filtered EEG, event markers, HR/RR, respiration, ALS, artifact channels, StimulusFingerprint regressors where available.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Slice by phase, compute sliding-window features, spectral bands, ROI summaries, artifact sentinels, autonomic summaries, and condition labels. Typically writes 1 s rows; OCM025 uses 0.25 s cue windows.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

*_time_resolved_feature_frame.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

This is the canonical physiology feature table. Visualizer *_features_used.csv is only the subset used for a render.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Missing feature table means most downstream modules cannot run.

Safe public wording

Recommended wording

This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

6. ArtifactScore

VETO AND COVARIATE LAYER

Tier: Primary gate Quantifies windows where EEG features may be contaminated by blink, jaw, motion, flatline, or high-frequency noise.

What the module is trying to prevent

This module exists primarily to prevent a false-positive interpretation. It asks whether the run, timing, stimulus, or signal quality is strong enough to allow any downstream claim. If this layer fails, the correct output is not a weaker positive result; it is a downgraded or non-interpretable run.

Inputs and assumptions

EEG channels, Fp1/Fp2/frontal channels, high-frequency sentinel bands, peak-to-peak amplitude.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Combine normalized peak-to-peak, line/high-frequency sentinel activity, ocular/frontal outlier features, and dropout/flatline checks. Penalize or veto contaminated windows.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

artifact_score columns, artifact tables, module veto flags.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

ArtifactScore has veto power. A gamma/theta event that is artifact-driven is not an endpoint.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

High artifact in candidate windows converts biological interpretation into artifact/confound caution.

Safe public wording

Recommended wording

This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

7. Canonical Analysis Frame and status-aware chain

IMPLEMENTED ANALYSIS CONTRACT

Creates one versioned table and manifest through which downstream modules receive canonical conditions, block instances, source columns, timing, QC, evidence grade, and missingness.

What the module is trying to prevent

Prevents differently shaped module inputs, silent row misalignment, recursive output selection, and the conversion of missing or failed evidence into favorable values.

Inputs and assumptions

- Recognized protocol and condition aliases.
- Unique contiguous block-instance identifiers.
- Monotonic time and declared source columns.
- Hard-QC, adapter grade, evidence grade, and missingness fields.

Algorithmic logic

Each module receives declared dependencies and returns an explicit state rather than a generic success/failure flag.

Outputs and where to find them

- Canonical analysis-frame tables.
- Authoritative chain manifest and artifact index.
- Per-module PASS, CAUTION, INELIGIBLE, NOT_APPLICABLE, or FAILED status.

How to interpret it

A module can be computationally successful yet scientifically cautioned. Exploratory troubleshooting after a failed primary gate cannot inherit confirmatory status.

How it can fail

Incorrect aliases, duplicate block IDs, ambiguous folders, or a module that ignores chain status can reintroduce silent errors.

Safe public wording

Recommended wording

The analysis chain completed under the declared dependencies and eligibility state; cautions and ineligible modules remain explicit.

8. Analysis-readiness preflight

PRE-COMPUTATION GATE

Checks whether a run is sufficiently described and covered for a requested analysis before the module starts.

What the module is trying to prevent

Prevents a numerical result from being produced merely because some similarly named columns or folders exist.

Inputs and assumptions

- Protocol and condition recognition.
- Baseline 1 availability.
- Timestamp gaps and block identity.
- Required features, coverage, artifact sentinels, ALS status, adapter grade, and optional ShotOrder/DGA/autonomic/anchor inputs.

Algorithmic logic

The preflight maps specific reasons to PASS, CAUTION, or INELIGIBLE. It does not convert optional absence into failure unless the requested endpoint requires that input.

Outputs and where to find them

- Human-readable readiness report.
- Machine-readable readiness JSON.
- Explicit missing and available feature inventory.

How to interpret it

PASS means the declared computation is supportable, not that the construct is valid or the result will be positive.

How it can fail

An overly permissive legacy adapter or mislabeled condition can produce a misleading PASS; adapter grade and reasons must be reviewed.

Safe public wording

Recommended wording

The run was computationally eligible for this module under the declared schema; readiness does not establish scientific validity.

PART II

Stimulus modeling and primary recognition-integration path

These modules model exogenous drive, candidate recognition timing, delayed integration, and the locked-anchor endpoint.

9. StimulusFingerprint

STIMULUS CHARACTERIZATION

Tier: Primary guardrail Extracts exogenous stimulus regressors from the final rendered media.

What the module is trying to prevent

This module exists primarily to prevent a false-positive interpretation. It asks whether the run, timing, stimulus, or signal quality is strong enough to allow any downstream claim. If this layer fails, the correct output is not a weaker positive result; it is a downgraded or non-interpretable run.

Inputs and assumptions

Control/Target/Override MP4s, cue schedule JSON, locked anchor JSON if available.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Compute luminance, visual change, histogram delta, cut events, flash proxy, audio RMS/envelope derivative, cue trains, cue phase, ALS/start-pulse regressors, and anchor stimulus vectors.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

qc/stimulus_exogenous_regressor_frame_all_conditions.csv, cet_regressors_all_conditions.csv, stimulus_rhythm_summary_all_conditions.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

This lets CET-R test whether physiology is explained by stimulus rhythm, cue rhythm, luminance, cuts, motion, or audio envelope.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Missing Target/Override regressors makes CET-R partial. Human QC is still required for control audio unintelligibility.

Safe public wording

Recommended wording

The stimulus-regressor check did or did not explain the observed pattern. This is a guardrail, not a proof of meaning.

10. CET - Cinematic Entrainment Tracking

STIMULUS-TRACKING DIAGNOSTIC

Tier: Primary guardrail / exploratory Estimates how much EEG/NIP-like features track external film/cue/audio/visual rhythm.

What the module is trying to prevent

This module exists primarily to prevent a false-positive interpretation. It asks whether the run, timing, stimulus, or signal quality is strong enough to allow any downstream claim. If this layer fails, the correct output is not a weaker positive result; it is a downgraded or non-interpretable run.

Inputs and assumptions

StimulusFingerprint regressors and feature table.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Fit stimulus-side regressors to physiology features within each branch. Evaluate train and blocked-cross- validation R2.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

cet_residualization_model_summary.csv, cet_cue_tracking_summary.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

High generalizable R2 means apparent effects may be stimulus tracking. Low/negative blocked-CV suggests stimulus regressors do not explain the module pattern.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Weak regressors, missing audio/video features, or overfit models prevent strong conclusions.

Safe public wording

Recommended wording

The stimulus-regressor check did or did not explain the observed pattern. This is a guardrail, not a proof of meaning.

11. CET-R - residualization against exogenous drive

ATTRIBUTION GATE

Tier: Primary guardrail Re-tests NIP/MRED/TTI after regressing out external media rhythms.

What the module is trying to prevent

This module exists primarily to prevent a false-positive interpretation. It asks whether the run, timing, stimulus, or signal quality is strong enough to allow any downstream claim. If this layer fails, the correct output is not a weaker positive result; it is a downgraded or non-interpretable run.

Inputs and assumptions

Feature table, CII/NIP/MRED metrics, StimulusFingerprint regressors.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Model features as functions of exogenous regressors, compute residuals, then recompute anchor integrals/contrasts.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

cet_residualized_cii_anchor_integrals.csv, residualized summaries.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

If Target remains greater than Control/Override after CET-R, the result is less likely to be reducible to raw stimulus rhythm.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

If effects vanish after residualization, meaning-specific interpretation weakens.

Safe public wording

Recommended wording

The stimulus-regressor check did or did not explain the observed pattern. This is a guardrail, not a proof of meaning.

12. GammaScalpel

ARTIFACT-SENSITIVE WORK-SIGNAL FAMILY

Tier: Feature extraction / secondary Separates lower-gamma candidate activity into narrower bands and context labels instead of treating gamma as meaning.

What the module is trying to prevent

This module supports the core analysis chain. It should be interpreted only in relation to timing, artifact, Control, Override, and stimulus-regressor checks.

Inputs and assumptions

EEG feature windows, ROI labels, artifact scores, condition labels.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Subdivide lower gamma, e.g. 30-35, 35-40, 40-45 Hz, compare Target/Control/Override, separate power from phase-locking, apply artifact and visual/task penalties.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

gammascapel_microband_phase_summary.csv, MeaningGamma/TaskGamma components.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Gamma is an artifact-sensitive candidate work-signal proxy, not a biomarker. It contributes to MR only if gates survive.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

EMG/ocular/artifact, cue/task load, or visual features can explain gamma-like changes.

Safe public wording

Recommended wording

This module contributed an artifact-sensitive recognition/work-signal proxy. It is not a standalone meaning biomarker.

13. Temporal Semantic Proxy (TSP)

CANDIDATE TIMING PROXY

Tier: Feature extraction / secondary A composite proxy for scene-timed semantic recognition pressure.

What the module is trying to prevent

This module supports the core analysis chain. It should be interpreted only in relation to timing, artifact, Control, Override, and stimulus-regressor checks.

Inputs and assumptions

MeaningGamma proxy, taper/peak features, artifact penalties, condition labels, anchor windows.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Score temporally localized recognition-like events using meaning-weighted gamma/TSP components, taper detection, and artifact-corrected normalization.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

TSP columns, TSP event candidates, visual overlays.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

TSP marks candidate semantic strike timing. It is not proof of comprehension or memory.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

TSP can be driven by visual salience, audio, task, or artifact if not controlled.

Safe public wording

Recommended wording

This module contributed an artifact-sensitive recognition/work-signal proxy. It is not a standalone meaning biomarker.

14. CandidateLocal KHT-topo

EXPLORATORY RECIPROCAL MODEL

Tier: Exploratory convergence Tests local reciprocal coupling between observed work-signal features and inferred network-state topology proxies.

What the module is trying to prevent

This module is designed to generate, constrain, or falsify hypotheses. Its job is not to certify PRAYCG. It helps diagnose mechanisms, artifacts, timing issues, burden states, and alternative explanations.

Inputs and assumptions

Feature table, candidate events, theta/meaning/TSP windows.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Fit event-local reciprocal loop models using recognition features and delayed integration/topology features; compute K_HT-topo and percentiles.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

candidate_local_kht_topo_mred_event_table.csv, kht visual overlays.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

K_HT-topo is a human-scale coupling proxy. It is not K_OSM and not cellular proof.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

High K without delayed theta/carryover or condition specificity is insufficient.

Safe public wording

Recommended wording

This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

15. MRED - Meaning Recognition / Encoding Dissociation

PRIMARY CONSTRUCT DECOMPOSITION

Tier: Primary conceptual endpoint Separates meaning recognition (MR) from encoding/integration (ENC).

What the module is trying to prevent

This module supports the core analysis chain. It should be interpreted only in relation to timing, artifact, Control, Override, and stimulus-regressor checks.

Inputs and assumptions

TSP, MeaningGamma, KHT-topo, theta handoff, topology/afterstate proxies, artifact and condition labels.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Compute MR and ENC scores at anchors/candidates. Classify quadrants such as MR_HIGH_ENC_HIGH or

MR_HIGH_ENC_LOW.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

mred_event_table.csv, candidate_local_kht_topo_mred_event_table.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

MRED is the conceptual core: recognition can appear without clear integration, and integration-like changes may be nonsemantic unless Target-specific.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

No Control/Override specificity, high artifact, or no delayed window weakens classification.

Safe public wording

Recommended wording

This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

16. A-MRED - anchor-locked MRED

PRIMARY CANDIDATE GATE

Tier: Boxed primary path Compresses the suite into a cleaner primary endpoint for future studies.

What the module is trying to prevent

This module supports the core analysis chain. It should be interpreted only in relation to timing, artifact, Control, Override, and stimulus-regressor checks.

Inputs and assumptions

Locked anchor JSON, feature table, MRED table, Control/Target/Override contrasts, QC gates, CET-R where available.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

For each predeclared anchor, require Target MR and ENC above thresholds and greater than Control and Override, with QC pass.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

amred_anchor_endpoint_table.csv, amred_primary_endpoint_summary.csv, amred_visual_overlay.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

A-MRED pass is the strongest current endpoint category. Strict use requires frame-verified locked anchors and clean timing.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Estimated anchors, weak ALS, missing Control/Override, or timing sensitivity produce pilot-positive, not confirmatory, results.

Safe public wording

Recommended wording

A predeclared anchor showed a Target-favoring candidate pattern under the specified gates. Strict confirmation still requires locked, frame-verified anchors, clean timing, artifact controls, and replication.

17. MRED-Peak

ACUTE CANDIDATE PROFILE

Tier: Boxed primary path / endpoint subtype Identifies acute anchor-locked meaning shock or recognition-plus-integration events.

What the module is trying to prevent

This module supports the core analysis chain. It should be interpreted only in relation to timing, artifact, Control, Override, and stimulus-regressor checks.

Inputs and assumptions

A-MRED, BIT, MRED scores, CII margins, KHT/MR/ENC tables.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Combine target MR, target ENC, strict gates, CII margins, and QC flags into a peak score and gate label.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

mred_peak_resolution_anchor_table.csv, top_mred_peak_candidates.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Best for Contact-style acute recognition events. A strict peak pass still requires anchor/timing confidence.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Peak candidates without strict lock stay exploratory.

Safe public wording

Recommended wording

A predeclared anchor showed a Target-favoring candidate pattern under the specified gates. Strict confirmation still requires locked, frame-verified anchors, clean timing, artifact controls, and replication.

18. MRED-Resolution

DELAYED CANDIDATE PROFILE

Tier: Secondary/endpoint subtype Identifies delayed reflective or regulatory recovery rather than a single acute peak.

What the module is trying to prevent

This module summarizes a pattern already estimated elsewhere. It is useful for comparing branches, runs, or anchor families, but it should not outrank the primary endpoint. It helps explain shape and direction, not final truth.

Inputs and assumptions

Baseline2 metrics, NUPI/RDI, EET, self-report echo, MRED/NIP scores.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Combine post-target recovery, TargetEchoedDuringBaseline2, EET afterstate echo, and RDI with anchor- level context.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

mred_peak_resolution_anchor_table.csv, top_mred_resolution_candidates.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Best for Field-of-Dreams-style closure/recovery. It is a profile, not proof of healing.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

No Baseline2 means true resolution cannot be graded; proxy-only results are flagged.

Safe public wording

Recommended wording

The run showed a recovery- or resolution-like profile in the available proxies. This is not a clinical or thermodynamic proof; it is an interpretable after-state profile.

PART III

Secondary summaries and contextual interpretation

These modules describe continuous immersion, task attenuation, update polarity, autonomic availability, and cumulative baseline change.

19. NIP - Narrative Immersion Proxy

SECONDARY SUMMARY

Tier: Secondary Translates attention + resonance into macroscopic PRAYCG proxies.

What the module is trying to prevent

This module summarizes a pattern already estimated elsewhere. It is useful for comparing branches, runs, or anchor families, but it should not outrank the primary endpoint. It helps explain shape and direction, not final truth.

Inputs and assumptions

MeaningGamma/TSP/NAST for semantic attention; theta/KHT/topology/API for integration; artifact/confounds.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Compute A_{sem} and R_{int} components; multiply/compose into immersion density.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

nip_component_timeseries.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

NIP estimates attention-plus-integration immersion proxy, not dopamine/oxytocin or literal immersion chemistry.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

High sensory drive or artifact can mimic attention; high recognition without integration should not be called full immersion.

Safe public wording

Recommended wording

This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

20. BIT - Bivariate Immersion Threshold

SECONDARY EVENT RULE

Tier: Secondary Ensures immersion is not a mere average of attention and resonance.

What the module is trying to prevent

This module exists primarily to prevent a false-positive interpretation. It asks whether the run, timing, stimulus, or signal quality is strong enough to allow any downstream claim. If this layer fails, the correct output is not a weaker positive result; it is a downgraded or non-interpretable run.

Inputs and assumptions

NIP components, MRED quadrants, artifact and condition gates.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Boolean gate requiring semantic attention and integration/resonance to exceed thresholds simultaneously, with condition specificity.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

bit_event_table.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

BIT pass approximates a strict MR_HIGH_ENC_HIGH immersion candidate.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

High attention but low integration fails BIT by design.

Safe public wording

Recommended wording

This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

21. CII - Continuous Immersion Index

SECONDARY ANCHOR INTEGRAL

Tier: Secondary Scores scene-level immersion density over anchor windows.

What the module is trying to prevent

This module summarizes a pattern already estimated elsewhere. It is useful for comparing branches, runs, or anchor families, but it should not outrank the primary endpoint. It helps explain shape and direction, not final truth.

Inputs and assumptions

NIP time series and anchor windows.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Integrate/average positive semantic-attention multiplied by integration proxy across a window, duration- normalized.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

cii_anchor_integrals.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Useful for sustained scenes, not just spikes. Longer windows must be normalized.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Can be inflated by duration, sensory rhythm, or artifacts if CET-R and QC are missing.

Safe public wording

Recommended wording

This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

22. IAQ / ITQ - Immersion Attenuation Quotient

SECONDARY TARGET-OVERRIDE SUMMARY

Tier: Secondary Estimates how much Override attenuates Target immersion proxy.

What the module is trying to prevent

This module summarizes a pattern already estimated elsewhere. It is useful for comparing branches, runs, or anchor families, but it should not outrank the primary endpoint. It helps explain shape and direction, not final truth.

Inputs and assumptions

Target and Override CII/NIP values for matched anchors.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Compute Target-minus-Override difference or ratio-bounded attenuation.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

iaq_target_override_table.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Positive IAQ means Target preserved more immersion proxy than Override. It is not a moral theft score.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

If Override equals/exceeds Target, reception/extraction claim weakens or story breaks through task.

Safe public wording

Recommended wording

This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

23. TTI - Reception-Extraction Tradeoff

SECONDARY TASK COMPARISON

Compares natural-viewing reception with analytic extraction without using literal thermodynamic language.

What the module is trying to prevent

Prevents an Override-related decrease in receptive measures from being interpreted as loss of meaning without accounting for greater task engagement and extraction load.

Inputs and assumptions

- Target and Override MR/ENC or compatible summaries.
- TaskGamma or extraction-load features.
- Artifact, cue, gaze, and confound burden.
- Matched media identity and branch coverage.

Algorithmic logic

The score combines Target-minus-Override recognition/integration and autonomic differences with Override-minus-Target task engagement, followed by penalties.

$$TTI = w1*z(MR_T-MR_O) + w2*z(ENC_T-ENC_O) + w3*z(API_T-API_O) + w4*z(Task_O-Task_T) - penalties$$

Outputs and where to find them

- Run-level reception-extraction summary.
- Anchor-level Target-Override contrasts.

- Component and warning tables.

How to interpret it

A positive value means the configured Target branch was more reception-like while Override was more extraction-like. It is not energy theft, moral loss, or literal thermodynamics.

How it can fail

Media mismatch, task noncompliance, poor cue visibility, order effects, or missing gaze data can dominate the contrast.

Safe public wording

Recommended wording

The Target-Override pattern was more consistent with receptive versus extraction-dominant processing under the configured operational measures.

24. NUPI - Narrative Update Polarity Index

SECONDARY UPDATE PROFILE

Tier: Secondary Classifies update profile as accommodative load versus resolute recovery.

What the module is trying to prevent

This module summarizes a pattern already estimated elsewhere. It is useful for comparing branches, runs, or anchor families, but it should not outrank the primary endpoint. It helps explain shape and direction, not final truth.

Inputs and assumptions

ALI components, RDI components, Baseline2, NUPI anchors, self-report echo, EET.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Compute ALI as semantic/update-load profile; compute RDI as regulatory-dividend/afterstate profile; NUPI

$$= \text{RDI} - \text{ALI}.$$

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

nupi_run_summary.csv, nupi_anchor_polarity_table.csv, nupi_visual_overlay.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Distinguishes high-load-with-recovery from resolute-recovery profiles. Does not measure heat/ATP/clinical healing.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

No Baseline2 or no afterstate physiology prevents reliable polarity grading.

Safe public wording

Recommended wording

The run showed a recovery- or resolution-like profile in the available proxies. This is not a clinical or thermodynamic proof; it is an interpretable after-state profile.

25. API-A - autonomic availability summary

SECONDARY PHYSIOLOGICAL SUMMARY

Tier: Secondary/covariate Summarizes regulated availability rather than simple arousal.

What the module is trying to prevent

This module summarizes a pattern already estimated elsewhere. It is useful for comparing branches, runs, or anchor families, but it should not outrank the primary endpoint. It helps explain shape and direction, not final truth.

Inputs and assumptions

HR, RMSSD/SDNN/pNN50, HR slope, respiration if available.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Weighted z-score composite favoring lower HR, higher HRV, stable/regulated trends, and respiratory context.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

api_a columns, autonomic summaries.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Helps distinguish receptive regulation from strain, but not a clinical measure.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Requires reliable R-R/ECG; respiration/common drive can explain HRV.

Safe public wording

Recommended wording

This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

26. Baseline 1 versus Baseline 2

CUMULATIVE RECOVERY COMPARISON

Tier: Secondary afterstate Compares pre-run baseline to final reflection baseline.

What the module is trying to prevent

This module summarizes a pattern already estimated elsewhere. It is useful for comparing branches, runs, or anchor families, but it should not outrank the primary endpoint. It helps explain shape and direction, not final truth.

Inputs and assumptions

Baseline phase features, HR/HRV, respiration, NIP/TSP/TaskGamma, artifact.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Compute feature deltas and regulation/reflective-state indicators.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

baseline1_vs_baseline2_summary.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Supports afterglow/recovery interpretation when Baseline2 is cleaner, more regulated, or semantically active.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Can be contaminated by fatigue, report burden, order effects, or artifact differences.

Safe public wording

Recommended wording

This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

27. DGA - Decoder Gate Availability

SECONDARY ATTRIBUTION MODEL

Separates semantic access, subjective availability, emotional/integrative impact, extraction load, and confound burden.

What the module is trying to prevent

Prevents poor comprehension, audio difficulty, or analytic task burden from being collapsed into a single narrative-effect interpretation.

Inputs and assumptions

- Branch reports for comprehension, volume, noise, and synchronization difficulty.
- Meaning, absorption, afterglow, story-active state, task load, and confound burden.
- Explicit normalization of report scales.

Algorithmic logic

A bounded gate combines positive access/availability/impact terms and subtracts extraction and confound load.

$$D_{gate,b} = sigmoid(2.0*SA_b + 1.4*DA_b + 1.1*EI_b - 1.2*X_b - 1.8*C_b - 0.75)$$

Outputs and where to find them

- Branch D_gate values.
- Target-Override Gate Dissociation Index.
- Labels describing open, split, unresolved, or absent availability.

How to interpret it

DGA explains why a primary result may pass, fail, or split. It does not replace A-MRED and is not a direct decoder of consciousness.

How it can fail

Self-report scaling, missing branch reports, familiarity, demand characteristics, or circular use of the same outcomes can distort the gate.

Safe public wording

Recommended wording
Decoder availability differed across branches under the declared access, impact, task, and confound inputs.

PART IV

Exploratory convergence and task-state modules

These modules test candidate state transitions, persistence, topology, algorithmic complexity, ocular unloading, and task microstates.

28. NAST - Narrative Absorption State Transition

EXPLORATORY TRANSITION MODEL

Tier: Exploratory / transition Detects branch-entry state transition into narrative absorption or task stance.

What the module is trying to prevent

This module is designed to generate, constrain, or falsify hypotheses. Its job is not to certify PRAYCG. It helps diagnose mechanisms, artifacts, timing issues, burden states, and alternative explanations.

Inputs and assumptions

Late washout windows, early branch windows, alpha, MeaningGamma, TSP, TaskGamma, theta.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Compare pre-branch and early-branch features; quantify meaning/alpha/task transitions.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

nast_phase_transition_table.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Useful for onset dynamics. Target may show meaning/TSP increase; Override may show task-gamma increase.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Alpha is eye/attention sensitive; NAST alone is not a meaning endpoint.

Safe public wording

Recommended wording

This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

29. Topo-OSM Network State

HUMAN-SCALE TOPOLOGY PROXY

Tier: Exploratory interpretation Represents human-scale latent state as network-state topology proxy.

What the module is trying to prevent

This module is designed to generate, constrain, or falsify hypotheses. Its job is not to certify PRAYCG. It helps diagnose mechanisms, artifacts, timing issues, burden states, and alternative explanations.

Inputs and assumptions

Theta, gamma, alpha, TSP, API-A, respiration, artifact, branch summaries.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Build phase/branch vectors and transition distances rather than cellular hidden-Y claims.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

topo_osm_phase_state_vectors.csv, topo_osm_visual_overlay.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Human-scale Topo-OSM only: state geometry, not microtubules or biophotons.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Too few channels, unconfirmed channel map, or artifact can make topology labels unreliable.

Safe public wording

Recommended wording

This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

30. EET - Endogenous Echo Tracking

EXPLORATORY PERSISTENCE MODEL

Tier: Exploratory convergence Tests whether scene-anchor state vectors resemble later washout/Baseline2 states.

What the module is trying to prevent

This module is designed to generate, constrain, or falsify hypotheses. Its job is not to certify PRAYCG. It helps diagnose mechanisms, artifacts, timing issues, burden states, and alternative explanations.

Inputs and assumptions

Anchor feature vectors, washout/Baseline2 feature vectors, NIP/MRED/EET tables.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Compute cosine/similarity or vector resemblance between anchor state and later afterstate windows.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

eet_endogenous_echo_tracking.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Suggests afterstate resemblance. It is not replay proof or memory proof.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Similarity may reflect baseline drift, common arousal, artifact, or generic state persistence.

Safe public wording

Recommended wording

The after-state resembled the anchor-state vector under the specified metric. This is not proof of replay or memory.

31. MRED-ITP - Information-Thermodynamic Proxy

EXPLORATORY PROXY FAMILY

Tier: Exploratory convergence Adds complexity and ocular-release proxies around MRED events.

What the module is trying to prevent

This module is designed to generate, constrain, or falsify hypotheses. Its job is not to certify PRAYCG. It helps diagnose mechanisms, artifacts, timing issues, burden states, and alternative explanations.

Inputs and assumptions

LZ-style complexity time series, blink/Fp1 proxy, MRED anchors, artifact.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Compute complexity strike/settlement and blink suppression/release metrics, then summarize convergence with MRED.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

lz_complexity_timeseries.csv, acg_event_table.csv, blink_event_table.csv, ocu_event_table.csv, mred_itp_anchor_summary.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

A convergence layer only. Complexity is not thermodynamic entropy; blink release is not memory encoding proof.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

High artifact, lack of EOG/eye tracking, or modality-specific complexity effects limit interpretation.

Safe public wording

Recommended wording

This module supplied convergence or falsification evidence. It does not prove entropy reduction or memory encoding.

32. ACG - Algorithmic Complexity Gate

EXPLORATORY COMPLEXITY DIAGNOSTIC

Tier: Exploratory Looks for complexity perturbation and settlement around anchors.

What the module is trying to prevent

This module is designed to generate, constrain, or falsify hypotheses. Its job is not to certify PRAYCG. It helps diagnose mechanisms, artifacts, timing issues, burden states, and alternative explanations.

Inputs and assumptions

Feature-state or raw EEG symbolic complexity windows.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Compare pre, peak, and post LZ-style complexity: strike = peak-pre; settle = post-peak.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

acg_event_table.csv, lz_complexity_timeseries.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Structured complexity trajectory can support perturbation/settlement interpretation.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Not literal entropy; artifact/noise can raise complexity; drowsiness/flatline can lower it.

Safe public wording

Recommended wording

This module supplied convergence or falsification evidence. It does not prove entropy reduction or memory encoding.

33. OCU - Ocular-Cognitive Unloading

EXPLORATORY OCULAR-STATE PROXY

Tier: Exploratory / ocular Tracks blink suppression and release around narrative anchors.

What the module is trying to prevent

This module is designed to generate, constrain, or falsify hypotheses. Its job is not to certify PRAYCG. It helps diagnose mechanisms, artifacts, timing issues, burden states, and alternative explanations.

Inputs and assumptions

Fp1/frontal ocular proxy or EOG if available, anchor windows, artifact table.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Detect blink-like events, compute blink rate before/during/after anchor, quantify suppression/release.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

blink_event_table.csv, ocu_event_table.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Possible event-boundary/attention-release marker. Stronger with EOG/eye tracking.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Fp1 proxy confounds blink with squint, EMG, eye strain, fatigue, or electrode motion.

Safe public wording

Recommended wording

This module supplied convergence or falsification evidence. It does not prove entropy reduction or memory encoding.

34. OCM025 - Override Cue Microstate

EXPLORATORY CUE ANALYSIS

Tier: Override diagnostic Analyzes digit recognition and working-memory updates around number cues at 0.25 s resolution.

What the module is trying to prevent

This module is designed to generate, constrain, or falsify hypotheses. Its job is not to certify PRAYCG. It helps diagnose mechanisms, artifacts, timing issues, burden states, and alternative explanations.

Inputs and assumptions

Cue schedule, raw/feature windows, Override/Target/Control comparisons.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Segment cue epochs into pre-cue, digit recognition, working-memory update, and maintenance windows; compare to nulls.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

ocm025_rsm_cvb_squint_cue_epoch_table.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Tests whether Override task was physiologically engaged. Not a meaning endpoint.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Poor cue visibility, missed cues, or artifacts can dominate.

Safe public wording

Recommended wording

This module describes task, cue, or visual-burden structure in Override. It does not prove a private mental strategy.

35. RSM - Running Sum Microstate Model

EXPLORATORY TASK-STATE MODEL

Tier: Override diagnostic Models arithmetic load, compute stall, deadline pressure, and latent guess risk.

What the module is trying to prevent

This module is designed to generate, constrain, or falsify hypotheses. Its job is not to certify PRAYCG. It helps diagnose mechanisms, artifacts, timing issues, burden states, and alternative explanations.

Inputs and assumptions

Cue values, running-sum trajectory, OCM025 features, task self-report.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Compute carry difficulty, high-carry load, hard-9 patterns, theta maintenance, closure delay, and latent guess-risk proxies.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

rsm_correlation_summary.csv, cue tables.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Identifies candidate burden/strategy windows, not proof of private strategy.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Requires accurate cue schedule and task report; cannot prove actual guessing.

Safe public wording

Recommended wording

This module describes task, cue, or visual-burden structure in Override. It does not prove a private mental strategy.

36. LSO / SPM - Subtitle Override and Phase Mapping

EXPLORATORY SUBTITLE CONTROL

Tier: Optional exploratory Analyzes subtitle-only or subtitle-mediated override variants.

What the module is trying to prevent

This module is designed to generate, constrain, or falsify hypotheses. Its job is not to certify PRAYCG. It helps diagnose mechanisms, artifacts, timing issues, burden states, and alternative explanations.

Inputs and assumptions

Subtitle file, bounding boxes, optional gaze, feature table, MRED/NIP outputs.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Compute lexical extraction cost, subtitle phase mapping, semantic timing shift, and subtitle MRED patterns.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

subtitle_line_event_table.csv, subtitle_phase_shift_table.csv, lso_visual_overlay.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Tests route cost: semantic access through text versus natural audiovisual absorption.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Without gaze, reading timing is estimated only. Audio+subtitle variants test a different condition.

Safe public wording

Recommended wording

This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

PART V

Expanded confound-defense modules

These modules challenge high-order stimulus structure, spatial attention, order, afterglow attribution, respiration, and task access.

37. CVB - Cue Visibility / Legibility Burden

CONFOUND DIAGNOSTIC

Tier: Override diagnostic Separates cue perception burden from arithmetic compute burden.

What the module is trying to prevent

This module is designed to generate, constrain, or falsify hypotheses. Its job is not to certify PRAYCG. It helps diagnose mechanisms, artifacts, timing issues, burden states, and alternative explanations.

Inputs and assumptions

Cue schedule, visual contrast/legibility metrics, OCM/RSM features, SquintProxy.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Combine weak digit-recognition, high holding burden, low contrast/blur reports, and visual-search proxies.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

cue legibility burden columns and candidate tables.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Flags possible hard-to-see cues. Particularly useful for Snack Attack and Her-like issues.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Cannot confirm invisibility without frame-level visibility and/or eye tracking.

Safe public wording

Recommended wording

This module describes task, cue, or visual-burden structure in Override. It does not prove a private mental strategy.

38. SquintProxy

CONFOUND DIAGNOSTIC

Tier: Artifact/covariate Uses frontal/Fp1-like artifact as possible visual strain sentinel.

What the module is trying to prevent

This module supports the core analysis chain. It should be interpreted only in relation to timing, artifact, Control, Override, and stimulus-regressor checks.

Inputs and assumptions

Fp1/frontal high-frequency, p2p, blink-like sentinels, self-report.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Score eye-strain/squint candidates separately from blink events and EEG endpoints.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

squint_proxy candidates within RSM/CVB tables.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Useful as covariate only. EOG/eye tracking is needed for confirmation.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Confounds with blink, EMG, brow tension, electrode movement, general effort.

Safe public wording

Recommended wording

This module describes task, cue, or visual-burden structure in Override. It does not prove a private mental strategy.

39. HOC-R - high-order confound residualization

ATTRIBUTION AUDIT

Tests whether candidate physiological or analysis features can be predicted from low-level and high-order properties of the stimulus.

What the module is trying to prevent

Prevents residual Target effects from being labeled narrative-specific when faces, bodies, objects, speech, motion, cues, or area-of-interest occupancy remain unmatched.

Inputs and assumptions

- Low-level luminance, audio envelope, cuts, optical flow, and cue regressors.
- Available face, body, object, speech, biological-motion, and AOI regressors.
- Canonical conditions including PhaseScrambled and ShotOrder where present.

Algorithmic logic

Fits declared models, reports collinearity and coverage, and prefers held-out behavior over training fit before residualization is interpreted.

```
y = X*beta + error
y_resid = y - y_hat
R2_train = 1 - sum((y-y_hat)^2)/sum((y-mean(y))^2)
```

Outputs and where to find them

- Regressor inventory and availability grade.
- Training and held-out model summaries where supportable.
- Residualized feature tables and explicit warnings.

How to interpret it

A residual constrains measured alternatives; it does not prove the residual is meaning. Missing high-order regressors leave Target-Phase cautioned.

How it can fail

Overfitting, collinearity, low event counts, missing annotations, or condition-specific regressor quality can make residualization misleading.

Safe public wording

Recommended wording

The candidate effect did or did not remain after the declared measured stimulus regressors; unmeasured confounds remain possible.

40. OSA - ocular and spatial-attention burden

EXPLORATORY CONFOUND DIAGNOSTIC

Summarizes cue duty, legibility, eye strain, visual competition, gaze/AOI burden, and task-related spatial attention.

What the module is trying to prevent

Prevents frontal, ocular, or gamma changes caused by cue monitoring and gaze behavior from being treated as narrative effects.

Inputs and assumptions

- Cue duration and interval.
- Legibility, blur, eye strain, stall, and task-load reports.
- Gaze or AOI data when available.

Algorithmic logic

Combines available cue-duty and report-burden components; missing gaze data triggers an explicit limitation.

```
cue_duty = display_duration/cue_interval
OSA = mean_available(cue_duty, report_burden)
```

Outputs and where to find them

- OSA summary and caution flag.
- Available-component inventory.
- Gaze/AOI limitation status.

How to interpret it

Higher OSA indicates greater configured ocular/spatial burden, not a probability that the EEG is ocular artifact.

How it can fail

Self-report may be incomplete, cue geometry may differ by display, and the current threshold is exploratory.

Safe public wording

Recommended wording

Ocular/spatial burden was elevated under the configured heuristic and should qualify task-related interpretation.

41. OHC - order, habituation, and carryover

WARNING HEURISTIC

Records branch order and evaluates whether fatigue, habituation, novelty loss, or prior-condition carryover could explain a contrast.

What the module is trying to prevent

Prevents a fixed-order difference from being interpreted as a pure condition effect or a short washout from being treated as a biological reset.

Inputs and assumptions

- Native and assigned sequence.
- Period, washout duration, branch reports, and Baseline 2.
- Participant/stimulus identifiers for prospective models.

Algorithmic logic

Produces explicit order/carryover warnings and preserves sequence/period for modeling. Prospective Williams variants reduce systematic first-order imbalance.

```
Observed(a-b) = condition(a-b) + period(a-b)
               + carryover_into_a - carryover_into_b + error
```

Outputs and where to find them

- Order and period table.
- Carryover/habituation warnings.
- Prospective sequence metadata.

How to interpret it

Counterbalancing improves balance but does not eliminate long-memory carryover or nonlinear order effects.

How it can fail

Small or incomplete sequence cells, reruns, fatigue, and stimulus-specific carryover can remain confounded.

Safe public wording

Recommended wording

The contrast is order-cautioned or order-balanced under the declared sequence; carryover remains a modeled limitation.

42. AAM - afterglow attribution model

EXPLORATORY ATTRIBUTION DIAGNOSTIC

Compares whether late state and Baseline 2 resemble Target afterglow or recovery/relief from the Override task.

What the module is trying to prevent

Prevents cumulative late physiology from being assigned uniquely to Target merely because a final reflection contains Target-related content.

Inputs and assumptions

- Baseline 2 features and branch-specific washout states.
- Target echo/afterglow and story-active reports.
- Override task load, compliance, stall, and recovery indicators.

Algorithmic logic

Contrasts a Narrative Afterglow Index with a Task-Completion/Relief Index.

```
NAI = mean_available(sim(B2,W2), target_echo, target_afterglow, story_active_W2)
TCRI = mean_available(task_load, compliance, stall, sim(B2,W3))
AAM = NAI - TCRI
```

Outputs and where to find them

- AAM score and component table.
- Target-like, task-relief-like, or mixed label.
- Cumulative-baseline warning.

How to interpret it

A positive exploratory score is Target-afterglow-like; a negative score is task-relief-like. The middle is mixed and Baseline 2 remains cumulative.

How it can fail

Order, long carryover, report demand, or weak branch-specific fingerprints can make attribution ambiguous.

Safe public wording

Recommended wording

The late profile was more Target-afterglow-like, task-relief-like, or mixed under the declared heuristic.

43. RespDualPath - physiology versus respiratory artifact

RESPIRATORY ALTERNATIVE EXPLANATION

Maintains separate physiological and artifact/confound interpretations of respiratory change.

What the module is trying to prevent

Prevents breathing-related movement or belt events from being treated automatically as meaningful state transitions.

Inputs and assumptions

- Respiration feature table or raw belt stream.
- Artifact overlap and EEG timing.
- Cardiac stream when coupling analysis is requested.

Algorithmic logic

The legacy path detects abrupt respiratory change; the continuous path estimates rate, depth, phase, velocity, acceleration, sigh/hold candidates, motion cautions, and HR-respiration coupling with time-shift nulls.

```
d_t = |resp_t-resp_(t-1)|  
legacy_threshold = mean(d) + 2.5*SD(d)
```

Outputs and where to find them

- Respiratory event tables.
- Physiology and artifact paths.
- Coupling/null tables and QC report.

How to interpret it

A respiratory event may be physiologically interesting, artifactual, both, or indeterminate. It remains separate from primary CAI.

How it can fail

Sensor inversion, belt slip, motion, arrhythmia, interpolation, and permissive hold rules can alter labels.

Safe public wording

Recommended wording

Respiration did or did not provide a plausible timing or artifact explanation under the declared path and QC rules.

PART VI

Current isolated exploratory candidates

These modules are implemented and software-tested but remain isolated from primary eligibility until prospective validation supports promotion.

44. Continuous Autonomic / RespDualPath

IMPLEMENTED EXPLORATORY MODULE

Produces continuous cardiac and respiratory arrays while preserving beat-event HRV and explicit interpolation flags.

What the module is trying to prevent

Prevents HRV from being calculated on interpolated display grids and separates physiological timing from respiratory artifact explanations.

Inputs and assumptions

- Retained cardiac beat/IBI events.
- Raw respiration stream with timestamps.
- Coverage, plausibility, gap, and motion QC.

Algorithmic logic

Builds a 4 Hz display/alignment grid, computes RMSSD/SDNN/pNN50 from retained beats, derives respiration phase/depth/rate and derivatives, and tests exploratory HR-respiration lag coupling with circular shifts.

```
HR_j = 60000/RR_j
RMSSD = sqrt(mean(diff(RR)^2))
p_shift = [1+count(|r_null|>=|r_obs|)]/(B+1)
```

Outputs and where to find them

- cardiac_beats.csv and cardiac_grid_4hz.csv.
- respiration_grid_4hz.csv and respiration_events.csv.
- autonomic_windows_1hz.csv, coupling nulls, QC JSON, and validation report.

How to interpret it

Interpolated HR is for alignment; beat-derived HRV remains the scientific quantity. Coupling is exploratory and multiplicity-sensitive.

How it can fail

Poor beat detection, respiratory inversion, gaps, motion, permissive hold rules, or a best-lag search can create unstable results.

Safe public wording

Recommended wording

Autonomic timing and respiratory dependence were estimated under the declared event, coverage, and null rules; they do not enter primary CAI.

45. CAI/SID - Controlled Access-Integration Index

FROZEN EXPLORATORY CANDIDATE

Combines positive fast access-related evidence, positive future slow-integration evidence, and a corrected positive-loop term into a bounded operational index.

What the module is trying to prevent

Prevents correlated jointly low signals, unestimable correlation, artifacts, task load, and invalid windows from being interpreted as favorable access-integration evidence.

Inputs and assumptions

- MeaningGamma/TSP/NIP-compatible fast features with recorded source adapters.
- ThetaIntegration/NAS/PosteriorAlpha-compatible slow features.
- Same-block future windows from 8-30 seconds.
- Artifact, HF, P2P, task, coverage, and Baseline 1 QC.

Algorithmic logic

Signals are transformed to E and Y in [0,1]. Positive excess above .5 is required for the product loop. Support is the equal-weight mean of positive E, positive future Y, and K; penalties reduce support. Hard-QC failures are missing, not zero.

```
e_plus=max(E-.5,0); y_plus=max(Y_future-.5,0)
K_product=2*sqrt(e_plus*y_plus)
K=.70*K_product+.30*max(corr(E,Y_future),0) [if estimable]
Support=mean(E_plus,Y_plus,K)
Penalty=mean(A_plus,X_plus)
CAI_core=Support*(1-Penalty)
SID_[a,b]=(1/(b-a))*integral_a^b CAI_core(t)dt
```

Outputs and where to find them

- Time-resolved CAI_core and components.
- Branch summaries, contrasts, occupancy/load, SID interval summaries.
- Warnings, validation/readiness report, code/config hashes.

How to interpret it

CAI is bounded but is not a probability. SID is an interval average of CAI. DGA/context and autonomic outputs remain separate sensitivity layers.

How it can fail

Optional-source adapters, short future coverage, within-run dependence, retrospective tuning, artifact leakage, or treating .50 as validated can invalidate interpretation.

Safe public wording

Recommended wording

CAI/SID was higher or lower under the frozen exploratory model; this does not establish consciousness or narrative reception.

46. Micro Handoff

SOFTWARE-VALID, CONSTRUCT-UNVALIDATED

Tests whether a positive local rise in temporal gamma is followed over fixed short lags by a positive rise in theta-integration activity.

What the module is trying to prevent

Prevents low-low similarity, stationary high-high activity, condition-specific best-lag selection, artifact windows, or output cadence from being mistaken for directional handoff.

Inputs and assumptions

- Raw EEG at a supportable sample rate and declared 16-channel map.
- Baseline 1 for robust normalization; no full-run fallback.
- T5/T6 temporal gamma and Pz/P3/P4/T5/T6 theta.
- Fixed equal-weight lag bank from 100-1000 ms.

Algorithmic logic

Uses a 25 ms output grid with 250 ms causal gamma and 500 ms causal theta windows. It separates state coordination C, directional positive handoff H, and activation-weighted density D. Hard artifacts and gaps are missing.

```
O(t)=clip01(max(F_100(t)-F_100(t-.1),0)/.25)
R(t,tau)=clip01(max(S_250(t+tau)-S_250(t+tau-.1),0)/.25)
H(t,tau)=sqrt(Q_t*Q_t+tau)*sqrt(O(t)*R(t,tau))
D(t,tau)=sqrt(Q_t*Q_t+tau)*mean(F,S_future)*(1-|F-S_future|)
```

Outputs and where to find them

- 25 ms feature and lag-series tables.
- Branch/contrast summaries.
- Circular-shift, phase-randomized, block-label, and artifact-matched checks.
- Readiness, QC, manifest, and human-readable report.

How to interpret it

Synthetic checks validate software behavior only. Historical directional handoff was mixed; density was descriptively higher in most eligible runs but is not directional evidence.

How it can fail

Unequal causal windows shift recovered lag, historical channels/rates vary, repeated self-runs are not independent, and no historical run passed both time-structure null families.

Safe public wording

Recommended wording

Micro Handoff did or did not exceed the declared nulls; it is not a consciousness detector or validated 40 Hz RPM gauge.

PART VII

Visualization, interpretation, and final reporting

These modules organize evidence for review without creating new scientific endpoints.

47. MasterSync Visualizer

AUDIT AND SYNCHRONIZATION DISPLAY

Tier: Visualization/audit Creates synchronized stimulus/data review videos with public/TSC or lab-diagnostic modes.

What the module is trying to prevent

This module supports the core analysis chain. It should be interpreted only in relation to timing, artifact, Control, Override, and stimulus-regressor checks.

Inputs and assumptions

Stimulus MP4, XDF or feature CSV, analysis output folder, overlay tables.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Slice branch by markers, repair timebases when possible, robust-scale panels, show missing-data labels, render event overlays/cards.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

analysis MP4, features_used.csv, events_used.csv, render_report.json.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Excellent for review and explanation. Does not certify endpoint validity.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Bad feature table, missing panels, or uncorrected event timebase can mislead.

Safe public wording

Recommended wording

This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

48. Offline Interpretive Report Generator

DETERMINISTIC EXPLANATION ONLY

Tier: Reporting / usability Generates a local rule-based plain-English report from a Master Suite output folder.

What the module is trying to prevent

This module supports the core analysis chain. It should be interpreted only in relation to timing, artifact, Control, Override, and stimulus-regressor checks.

Inputs and assumptions

Analysis folder tables, optional auto-run MRED-Peak/Resolution.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Detect available tables, summarize QC, A-MRED, MRED-Peak/Resolution, NIP/TTI/NUPI/CET/EET/ITP/OCM/RSM, and boundaries.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

offline_interpretive_report.md/txt/json.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Helps GitHub/OSF users interpret outputs without uploading data to an AI assistant.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Rule-based, not flexible scientific reasoning. It cannot rescue bad inputs or discover unmodeled confounds.

Safe public wording

Recommended wording

This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

Final recommended interpretation workflow ☐ Start with run identity, media hashes, and Target/Override hash match.

☐ Check channel map, stream inventory, phase coverage, and ALS/display timing.

☐ Check artifact and confound tables before looking at gamma or theta.

☐ Check StimulusFingerprint/CET-R to see whether movie rhythm or cue rhythm explains the pattern.

- ☐ Use A-MRED/MRED-Peak/Resolution as the primary interpretive compression.
- ☐ Use NIP/TTI/NUPI as secondary summaries of immersion, extraction, and polarity.
- ☐ Use KHT-topo, NAST, EET, MRED-ITP, OCM/RSM/CVB/Squint as convergence or falsification layers.
- ☐ Read self-report as context and a consistency check, not proof.
- ☐ If the offline interpreter gives a positive summary but the QC layer failed, the QC layer wins.

Glossary of claim grades
Grade Meaning
Strict confirmatory candidate Locked, frame-verified anchors; clean timing; artifact/CET-R pass; Target > Control and Target > Override; preregistered rule.

Runner-registered estimated pilot Anchor file loaded before acquisition, but times are estimated or not frame-verified. Useful but not confirmatory.

Conceptually predeclared exploratory Anchor family existed before analysis, but was not loaded into the runner. Stronger than pure fishing, weaker than formal registration.

Exploratory discovery Found after analysis. Useful for future-freezing only.

Cautioned / non-interpretable Timing, artifact, stimulus, channel-map, or confound problem prevents strong interpretation.

49. Final interpretation workflow

REQUIRED REVIEW SEQUENCE

A defensible review moves from identity and timing to eligibility, primary candidates, secondary context, confound defenses, exploratory convergence, and independent outcomes.

Order	Review question
1	Are run identity, media hashes, protocol, condition markers, channels, and timebase correct?
2	Did hard timing, coverage, artifact, and applicability gates pass?
3	What did the predeclared primary candidate show at locked anchors?
4	Do secondary summaries explain task, access, autonomic, or baseline context?
5	Do HOC-R, CET-R, OSA, OHC, AAM, RespDualPath, and DGA offer simpler explanations?
6	Do exploratory modules converge without contradicting nulls or eligibility?
7	Do independent recognition, comprehension, confidence, narrative-coherence, and reports support the interpretation?

Final rule

No visualization, interpreter sentence, composite score, or exploratory module can rescue failed acquisition, missing required inputs, an ineligible primary endpoint, or contradictory independent evidence.

Claim-grade glossary

Grade	Meaning
STRICT_CONFIRMATORY	Predeclared, machine-registered, frame/timing verified, quality gates passed, and prospectively analyzed.
RUNNER_REGISTERED_ESTIMATED	Registered before collection but dependent on estimated timing or unverified frames.
CONCEPTUALLY_PREDECLARED	Specified in planning but not machine registered in the run.
EXPLORATORY	Developed, discovered, tuned, or selected after physiology review.
INELIGIBLE	Required evidence or gates are absent; the endpoint must not be interpreted for that run.